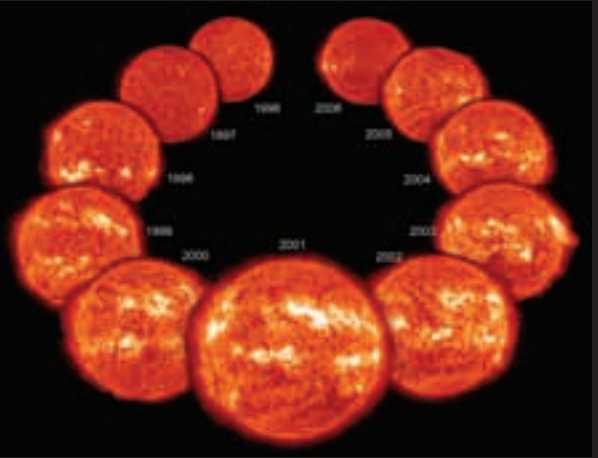




Left: The International Space Station will be at risk from the sun | Right: Even the sun has a climate cycle – a season is about 11 years



All this magnetic induced turbulence causes electrons and plasma to mix into a deadly cocktail that eventually explodes with the power of a hundred million Hiroshima's. Plasma is sent shooting out of the surface of the sun, several lakh kilometres above and then gets pulled back. However, sometimes these explosions are supermassive and end up actually travelling away from the surface of the sun into space. These are called coronal mass ejections (CMEs) and at times contain nearly a third of the Earth's mass!

Of course, most often these CMEs send plasma out into space away from the Earth, but often enough, they send plasma hurtling straight for us at 1,000 Km per second!

That's when scientists and astronomers get a little hot under the collar.

KILLER PLASMA FROM OUTER SPACE

Although Earth's own magnetic field is quite capable of handling the sun's regular tantrums, there are two matters that concern scientists.

First, if the sun were to spit venom directly at us in the form of a superstorm-caused CME, all of our technology could be in serious danger. Electromotive forces could cripple everything that's not secured in a Faraday box.

Secondly, the Earth has been known to swap magnetic poles every few hun-

dred thousand years, and the last flip was roughly 780,000 years ago – we're long overdue.

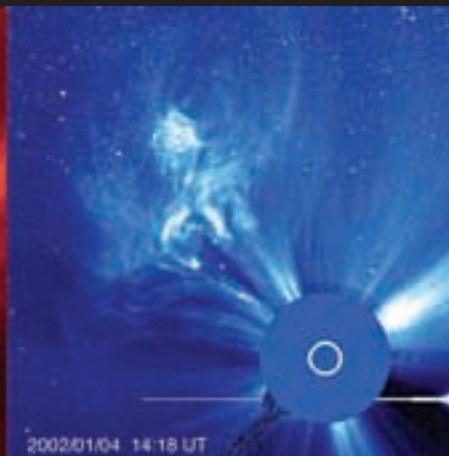
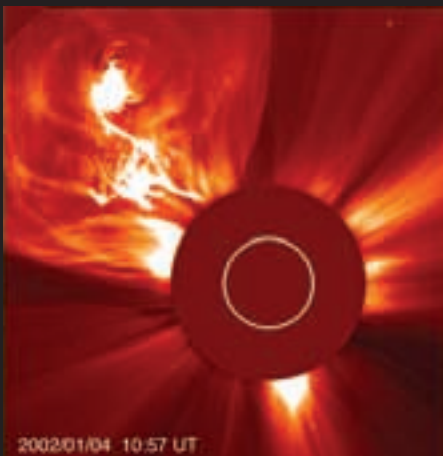
THE REAL GLOBAL WARMING

The problem with the Earth's polar switch is that it will disrupt the magnetic field and leave the Earth relatively unprotected from the sun's fury for anywhere between 4,000 to 7,000 years. Looking at ancient lava flows, scientists have discovered that during this period, although the Earth's magnetic field does drop, it doesn't drop to zero.

Still, 4,000 years with reduced protection against the solar wind will play havoc with life on Earth. Birds will not know which way is south, and die off. Increased solar effects reaching us will mean increased radiation, and as a result more deaths from cancer. Satellite communication will become near impossible, and since the atmosphere will expand due to the heat, we might even find it harder to breathe – just like when you climb a mountain.

We can forget about our Android phones and iPads; because electricity will be scarce. If you can't bear the heat or cold during a power cut, and get bored in 15 minutes without a gadget, think about what our civilisation will endure if most of it had to go a few thousand years without electricity? 

SOHO/EIT and SOHO/LASCO (ESA & NASA)



Solar flare captured on an ultraviolet sensor, an angle and Spectrometric Chronograph

TIMING MATTERS

A study suggests that timing is important when thinking of conceiving a baby. Apparently, those conceived and born during a peak in the solar cycle, on average, live 1.5 years less than those conceived and born during a lull in solar activity.

RECESSION AND THE SUN

Another study suggests that the stock market closely follows solar activity. When the activity in a solar cycle is increasing, so do stock market indexes, and when there's a lull, well you get 2008 nightmares. An interesting read: <http://bit.ly/f6qUgk>

CORONAL MASS EJECTIONS

Imagine a giant geyser spewing outwards, millions of miles into space, and at temperatures that form ionised plasma. That's what happens on the surface of the sun all too often. One such large CME caused the Solar Superstorm of 1859.